

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0607

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO5: Introduction – NumPy Arrays – NumPy Basics- Math – Random – Indexing – Filtering – Statistics – Aggregation – saving data – Data analysis with Pandas – DataFrame – Combining – Indexing – File I/O – Grouping – Filtering – Sorting – Plotting (BL 3)

Assessment Tool Weightage: 20%

QUESTIONS:03

Total Marks:25

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

I VIMALA K 192424276 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

Criteria	Conceptual Understanding	Accuracy of Answers	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total
Weightage	25%	25%	20%	15%	15%	
Q1						
Q2						
Q3						

Total Marks :

Annexure A

1. Create a pseudocode algorithm to detect and handle **outliers in a dataset** using statistical methods such as Z-score or IQR. The algorithm should identify extreme values, decide whether to remove or cap them, and ensure that the resulting dataset maintains consistency and reliability for further analysis.
2. Write a pseudocode algorithm to implement **data aggregation and summarization** across multiple categories. The algorithm should group data based on specified attributes, compute aggregated metrics (sum, average, count), and generate a summarized report. Include steps for sorting, indexing, and validating the correctness of aggregated results.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Pseudocode Algorithm for Outlier Detection and Handling

Aim:

To develop a pseudocode algorithm for detecting and handling outliers in a dataset using statistical methods such as Z-score and IQR, while maintaining the consistency and reliability of the dataset.

Pseudocode:

Input: Dataset D

Output: Cleaned Dataset D

Step 1: Start

Step 2: Read the dataset D.

Step 3: Select all numerical attributes from D.

Step 4: For each numerical attribute X in D, perform the following:

Step 4.1: Calculate the first quartile Q1.

Step 4.2: Calculate the third quartile Q3.

Step 4.3: Calculate the Interquartile Range:

$$\text{IQR} = Q3 - Q1$$

Step 4.4: Calculate the lower and upper limits:

$$\text{Lower Limit} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Limit} = Q3 + 1.5 \times \text{IQR}$$

Step 4.5: Identify values where:

$$X < \text{Lower Limit} \text{ OR } X > \text{Upper Limit}$$

Step 4.6: Mark these values as outliers.

Step 5: Calculate the percentage of outliers in the dataset.

Step 6: For each detected outlier:

IF the outlier is an incorrect or invalid value

THEN

Remove the corresponding record.

ELSE

Cap the value within the valid range:

IF $X < \text{Lower Limit}$

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        Set X = Lower Limit
    ELSE IF X > Upper Limit
        Set X = Upper Limit
    END IF
END IF
```

Step 7: Recalculate the mean, median, minimum, and maximum values after handling the outliers.

Step 8: Verify that the cleaned dataset does not contain unacceptable extreme values.

Step 9: Check the dataset for missing and inconsistent values.

Step 10: Compare the original and cleaned datasets to ensure that important information has not been unnecessarily removed.

Step 11: Save the cleaned dataset for further analysis.

Step 12: Stop.

Z-Score Method:

Input: Numerical Dataset D

Output: Cleaned Dataset D

Step 1: Start

Step 2: Calculate the mean and standard deviation of the dataset.

Step 3: For each value X, calculate:

$$Z = (X - \text{Mean}) / \text{Standard Deviation}$$

Step 4: IF $|Z| > 3$

THEN

Mark X as an outlier.

ELSE

Keep X unchanged.

END IF

Step 5: Remove obvious data-entry errors.

Step 6: Cap valid extreme observations when necessary.

Step 7: Validate the cleaned dataset.

Step 8: Save the cleaned dataset.

Step 9: Stop.

Result:

The algorithm successfully identifies extreme values using IQR or Z-score, removes incorrect outliers or caps valid extreme values, and produces a consistent, reliable, and analysis-ready dataset while preserving maximum data integrity.

2. Pseudocode Algorithm for Data Aggregation and Summarization

Aim:

To develop a pseudocode algorithm for grouping data across multiple categories, calculating sum, average, and count, and generating a summarized report with proper sorting, indexing, and validation.

Pseudocode:

Input: Dataset D

Output: Summarized Report R

Step 1: Start

Step 2: Read the dataset D.

Step 3: Identify the required grouping attributes
such as Category, Department, Region, or Product.

Step 4: Identify the numerical attribute required
for aggregation.

Step 5: Check the dataset for missing or invalid values.

Step 6: Sort the dataset based on the selected
grouping attributes.

Step 7: Create an index for the grouping attributes
to enable faster access and processing.

Step 8: Group the records according to the selected
category or combination of categories.

Step 9: For each group, calculate:

- a) $\text{Count} = \text{Number of records in the group}$
- b) $\text{Sum} = \text{Sum of numerical values in the group}$
- c) $\text{Average} = \text{Sum} / \text{Count}$

Step 10: Store the calculated results for each group
in the summarized dataset R.

Step 11: Sort the summarized results based on the
required category or aggregated metric.

Step 12: Generate the summarized report containing:

Grouping Category

Count

Sum

Average

Step 13: Validate the aggregated results:

a) Check whether the total count in all groups equals the number of records in the original dataset.

b) Check whether the sum of all group totals equals the overall sum of the original dataset.

c) Verify that the calculated averages are:

$$\text{Average} = \text{Sum} / \text{Count}$$

Step 14: If validation is successful,

display and save the summarized report.

Step 15: Otherwise, identify the incorrect group

and repeat the aggregation process.

Step 16: Stop.

Example of Summarized Report:

Category	Count	Sum	Average
Category A	10	750	75.0
Category B	8	640	80.0
Category C	12	900	75.0
Total	30	2290	76.33

Result:

The algorithm successfully groups the dataset based on specified attributes, calculates count, sum, and average for each category, sorts and indexes the data for efficient processing, and validates the aggregated results before generating the final summarized report.